12 · Multiple alleles, blood groups and X-linked genes

The words of this track, as it says them. Bold brown is the examiner.

Multiple alleles, blood groups, and X-linked genes.

Briefly explain multiple allele inheritance and sex-linked inheritance, using the right genetics terms.

Two more of the inheritance patterns on her checklist: multiple alleles, and X-linked genes. Her quizzes turn both into scenario questions, so it's worth having the crosses ready.

The hook: blood groups have three alleles, and a boy's X has no partner.

Multiple allele inheritance: a trait controlled by more than two alleles. Her example is the ABO blood group. The gene comes in three versions, A, B and O, and you carry two of them, so there are six genotypes. Genotypes AA and AO both give group A. BB and BO both give group B. AB gives group AB. And OO gives group O. The A and B alleles are codominant, and both are dominant over O.

The crosses she asks. AO crossed with BO gives all four groups, A, B, AB and O, twenty-five per cent each, which is how one family can have four blood groups. An O child needs an O from each parent. A man who is OO, whose children are all AO, must have a partner who is AA. And a man who is homozygous for group A, crossed with a woman who is heterozygous for group B, AA by BO, gives fifty per cent AB and fifty per cent AO.

The trap she sets: polygenic is not the same as multiple alleles. Polygenic means many genes. Multiple alleles means one gene with more than two versions. Her true or false says blood groups are polygenic. False.

X-linked, or sex-linked, inheritance. The gene sits on the X chromosome. A male has one X and a Y, and the Y has no matching allele to mask it. So one copy of a recessive X-linked allele always shows in a male, while a female needs two. That's why males inherit more sex-linked conditions, and why only females can be carriers. Her drop-down: males cannot be carriers, because they only carry one X. Her examples are colour blindness and haemophilia.

Her two crosses. A carrier mother, X big B, X little b, and a father with normal vision, X big B, Y: each son has a fifty per cent chance of being colour blind, and none of the daughters will be, though half of them will be carriers. A colour-blind father, X little b, Y, and a mother with normal vision, X big B, X big B: none of the sons are colour blind, because they get his Y, and all of the daughters are carriers with normal vision.

Once more. Multiple alleles, and X-linked genes.

Multiple alleles: one trait with more than two alleles, like ABO blood: three alleles, six genotypes, four groups. The A and B alleles are codominant and both dominant over O, so AO crossed with BO gives A, B, AB and O, twenty-five per cent each. It isn't polygenic.

X-linked: the gene is on the X and the Y has no matching allele, so one copy shows in a male while a female needs two, and only females can be carriers. A carrier mother and a normal father: half the sons affected, and no daughters. An affected father and a normal mother: no affected sons, and every daughter a carrier.